LCP Lens Correction Converter

v1.3

Overview & User Manual

Converts Adobe LCP distortion and vignette calibration data into lensfun / darktable-compatible PTLens and PA model XML.

prompted by John L. Nevill · coded by Claude

1. Overview

The LCP Lens Correction Converter is a self-contained, single-file HTML application that runs entirely in the browser — no server, no installation, no build step required. It reads an Adobe Lens Correction Profile (`.lcp`) file and converts its embedded distortion and vignetting calibration data into the two formats consumed by lensfun and, by extension, darktable.

1.1 The Problem It Solves

Adobe stores lens correction data in its ACM (Adobe Camera Model) format. Darktable's Lens Correction module uses lensfun, which requires two distinct polynomial representations:

- **PTLens model** — geometric distortion coefficients `a`, `b`, `c`
- **PA (Polynomial Approximation) model** — vignetting coefficients `k1`, `k2`, `k3`

These representations differ from ACM in normalisation radius, polynomial basis, and coordinate convention. The tool performs the mathematical translation analytically and precisely, producing ready-to-use lensfun XML.

1.2 Outputs Produced

- **Distortion conversion:** ACM radial `k1/k2/k3` → PTLens `a/b/c` via analytical least-squares (Cramér's rule)
- **Vignetting conversion:** ACM `α1/α2/α3` → lensfun PA `k1/k2/k3` via exact normalisation rescaling
- **Export .txt:** Annotated plain-text report with full conversion methodology and all coefficients
- **Export lensfun .xml:** Complete `<lensdatabase>` XML fragment for direct use in a lensfun database file

1.3 Attribution

The header of the tool reads: *'prompted by John L. Nevill · coded by Claude'*.

2. Sensor Presets and Geometry Constants

Selecting a sensor sets the physical dimensions used in every normalisation calculation. Six presets are built in:

Key	Name	Dimensions (mm)	Default Mount	Crop Factor
x2d	Hasselblad X / GFX	43.8 × 32.9	Hasselblad X	~ 0.79
ff	Full Frame	36.0 × 24.0	Nikon F	1.0
apsc	Fujifilm APS-C	23.8 × 15.6	Fujifilm X	~ 1.53
capsc	Canon APS-C	22.2 × 14.8	Canon EF-S	~ 1.6
mft	Micro Four Thirds	17.3 × 13.0	Micro Four Thirds	2.0
sony1	Sony 1″	13.2 × 8.8	Sony E	~ 2.7

2.1 Derived Geometry Constants

Three constants are derived from the selected sensor dimensions and govern all polynomial conversions:

Constant	Definition	Used for
`R_NORM`	Half the short side of the sensor (mm)	Lensfun r = 1 normalisation for PTLens distortion model
`R_MAX`	Half-diagonal of the sensor (mm)	Lensfun normalisation for vignetting (PA model); aperture-invariant
`r_max`	R_MAX ÷ R_NORM	Maximum radius in PTLens coordinate space; clamps the fit domain

Important: The sensor selection must match the camera body the LCP was calibrated for. The vignetting PA conversion uses R_MAX from

the selected sensor. A mismatch produces incorrect vignetting coefficients.

3. Conversion Mathematics

3.1 Distortion: ACM → PTLens

The two distortion residual polynomials being reconciled have incompatible basis functions. ACM uses odd powers; PTLens uses consecutive powers:

```
ACM residual:    Δ(r) = K1·r³ + K2·r⁵ + K3·r⁷

PTLens residual: Δ(r) = c·r²  + b·r³  + a·r⁴
```

where $K1 = k1 \cdot s^2$, $K2 = k2 \cdot s^4$, $K3 = k3 \cdot s^6$, **and** $s = R_NORM / focal_mm$.

The tool performs an analytical least-squares fit of the PTLens polynomial to the ACM polynomial over the integration domain $[0, rm]$**, solving the 3×3 normal equations in closed form using Cramér's rule. The upper limit is:**

```
rm = min(R_FIT_MAX,  focal_mm / R_NORM)
```

This clamp enforces the ACM valid range ($u \leq 1$ **in focal-length space). No numerical optimisation is involved; the result is the exact least-squares solution for the chosen domain.**

Note: ACM uses odd-power bases {r^3, r^5, r^7} while PTLens uses {r^2, r^3, r^4}. These are fundamentally different polynomial families. For lenses with large distortion the corner fit may diverge slightly. The distortion graph visualises this.

3.2 Vignetting: ACM → Lensfun PA

The vignetting conversion is an exact change of normalisation variable rather than a fitting operation. ACM normalises the polynomial radius to focal length; lensfun normalises to the sensor half-diagonal (`R_MAX`). The rescaling is:

```
k1_pa = α1 × (R_MAX / focal)²

k2_pa = α2 × (R_MAX / focal)⁴

k3_pa = α3 × (R_MAX / focal)⁶
```

No information is lost. The illumination model applied at render time is:

```
L(r) = 1 + k1·r² + k2·r⁴ + k3·r⁶   (r normalised to R_MAX)

correction_factor = 1 / L(r)
```

Vignetting data is exported for all apertures and all focus distances in the LCP regardless of the filter settings, because lensfun interpolates across the full vignetting table at render time.

4. User Interface Walkthrough

4.1 Loading a File

Drag-and-drop a `.lcp` or `.xml` file onto the drop zone, or click the zone to open a file browser. A pulsing filename badge confirms the file is loaded. The status bar reports the total number of calibration entries found and lists all aperture values present in the file.

Note: LCP files are XML/RDF documents conforming to Adobe's stCamera namespace. The parser extracts LensPrettyName, Make, Lens, SensorFormatFactor, and all calibration entry nodes without any server-side processing.

4.2 Sensor Selection

Click one of the six sensor buttons to set the physical dimensions governing all normalisation calculations. The sensor strip displays the live-computed values: sensor name and dimensions, diagonal in mm, `R_NORM`, and `r_max`.

The Mount field is an editable text input pre-filled from the sensor preset. Its value writes directly into the lensfun XML `<mount>` element. Edit it for adapted lenses or non-standard mount configurations.

4.2.1 Sensor Mismatch Warning

If the LCP's `SensorFormatFactor` attribute does not match the selected sensor's expected crop factor (within a tolerance window), an amber warning banner appears:

```
⚠  Sensor mismatch: LCP SensorFormatFactor X.XX — selected sensor diagonal YY.YY mm.
Vignette PA conversion uses R_MAX from the selected sensor.
```

Correct the sensor selection before exporting. Distortion coefficients are unaffected by sensor mismatch; only the vignetting PA rescaling uses R_MAX.

4.3 Filter Controls

A green information banner reminds you that darktable implements only one aperture and focus distance pair for lensfun distortion. The default filter values are therefore the recommended ones for most use cases.

4.3.1 Aperture Filter

The aperture dropdown is populated from all `ApertureValue` entries found in the LCP and defaults to f/8 if present, otherwise 'All apertures'. Selecting a specific aperture filters the distortion table and the `.txt` export.

4.3.2 Focus Distance Filter

The focus distance dropdown is populated from all `FocusDistance` values and defaults to the maximum available (infinity, stored as 10000 m in LCP files). Filtering by distance narrows the distortion table accordingly.

Note: Vignetting is always exported unfiltered — all apertures and distances regardless of the filter settings. Lensfun requires the full table to interpolate at render time.

4.4 Results Tables

Two collapsible tables appear after a file is loaded. Click the header row of either table to expand or collapse it. Summary chips above the tables show the active focal lengths, apertures, focus distances, R_NORM, and r_max.

4.4.1 Distortion Table — Filtered Entries

Column group	Columns	Description
Entry	#, FocalLength, FocusDistance, Aperture	Calibration entry parameters from the LCP
ACM radial (green)	k1, k2, k3	RadialDistortParam1/2/3 from the LCP PerspectiveModel element
PTLens (blue)	a, b, c	Converted PTLens coefficients in lensfun convention

4.4.2 Vignetting Table — All Entries

Column group	Columns	Description
Entry	#, FocalLength, Aperture, Distance	Calibration entry parameters from the LCP
ACM (α1α2α3) (green)	α1, α2, α3	VignetteModelParam1/2/3 from the LCP
PA k1 k2 k3 (blue)	k1, k2, k3	Converted lensfun PA vignetting coefficients

4.5 Interactive Graphs

Only one visualisation panel is open at a time. Opening any panel closes the others automatically.

4.5.1 Distortion Graph

Plots distortion correction percentage vs. radius in mm along the sensor diagonal (centre to corner). For each filtered entry, the solid line shows the ACM curve and the dashed line shows the PTLens conversion. Multiple focal lengths receive distinct colour pairs. Hover to read precise percentage values at any radius.

4.5.2 Vignette Graph

Plots illumination increase in EV (the correction applied at render time) vs. radius in mm, using the maximum available focus distance. One curve per aperture. The

0 EV dashed baseline indicates no correction. Hover to read per-aperture EV values. The graph plots `−log₂(L(r))` in EV.

5. Export

5.1 Export .txt

Downloads a fully self-documenting plain-text report named `{lensname}_{aperture}_{focusdist}.txt`. Contents:

- Source filename, export timestamp, entry counts
- Active filter settings (aperture and focus distance)
- Sensor geometry: name, dimensions, diagonal, `R_NORM`, `R_MAX`, `r_max`
- Full conversion methodology: polynomial forms, integration domain, normalisation convention
- Columnar data table with all filtered distortion entries and their PTLens equivalents
- Ready-to-paste `<distortion>` XML snippets for PTLens model
- All `<vignetting>` XML snippets for PA model (all apertures and distances)
- Column definitions and coefficient conventions

5.2 Export Lensfun .xml

Downloads a complete, paste-ready lensfun XML file named `{lensname}_lensfun.xml`. The file is immediately usable as a lensfun database entry:

```
<lensdatabase>


  <lens> <!-- John L Nevill 2025 -->

    <maker>...</maker>

    <model>...</model>

    <mount>...</mount>

    <cropfactor>...</cropfactor>

    <calibration>


      <!-- Distortion -->

      <distortion model="ptlens" focal="56" a="-0.001234" b="0.005678" c="-0.002345" /
>
```

```
      <!-- Vignetting -->
      <vignetting model="pa" focal="56" aperture="1.2" distance="10000"
                  k1="-0.2340" k2="0.0560" k3="-0.0120" />
    </calibration>
  </lens>


</lensdatabase>
```

The `<maker>`, `<model>`, and `<cropfactor>` fields are populated from LCP metadata. The `<mount>` value is taken from the editable Mount field in the sensor strip.

6. The Lensfun Database: Structure and Integration

Before integrating exported XML into lensfun, it is essential to understand how the database is organised. Incorrect integration — wrong mount name, duplicate entries, mismatched cropfactor, or data placed in the wrong file — will cause the lens either not to appear in darktable, or to apply corrections to the wrong camera-lens combination.

6.1 System Database vs User Database

Lensfun maintains two distinct database tiers:

Tier	Location (Linux)	Purpose	Who edits it
System	/usr/share/lensfun/version_1/	Official community-maintained lens data shipped with the lensfun package	Package manager only — never edit directly
User	~/.local/share/lensfun/	Personal additions and overrides; takes precedence over system data when lens identifiers match	You

Typical user database paths by operating system:

- **Linux:** `~/.local/share/lensfun/`
- **macOS:** `~/Library/Application Support/lensfun/db/`
- **Windows:** `%LOCALAPPDATA%\lensfun\db\`

If the directory does not exist, create it. Lensfun scans all XML files in the directory at startup.

Important: Never edit files in the system database directory. Package updates will silently overwrite your changes. All custom entries belong in the user database.

6.2 Database File Structure

Each XML file in the database contains one root `<lensdatabase version="1">` element enclosing any number of `<lens>` entries. The system database organises files by manufacturer (e.g. `fujifilm.xml`, `sony-e.xml`). In the user database, any valid filename works; a common convention is to name files after the lens:

```
~/.local/share/lensfun/fujifilm-xf56mmf12.xml
```

A minimal valid file looks like this:

```
<lensdatabase version="1">
  <lens>
    <maker>Fujifilm</maker>
    <model>XF 56mm f/1.2 R</model>
    <mount>Fujifilm X</mount>
    <cropfactor>1.53</cropfactor>
    <calibration>
      <distortion model="ptlens" focal="56" a="-0.001" b="0.004" c="-0.002" />
      <vignetting model="pa" focal="56" aperture="1.2" distance="10000"
                  k1="-0.23" k2="0.05" k3="-0.01" />
    </calibration>
  </lens>
</lensdatabase>
```

6.3 Anatomy of a Lens Entry

Every `<lens>` element requires the following child elements. Lensfun uses these to identify and match the lens when a raw file is opened in darktable:

Element	Required	Notes
`<maker>`	Yes	Manufacturer name. Must match an existing maker string in the lensfun database exactly — check the system database files for the correct spelling for your manufacturer.
`<model>`	Yes	Lens model string. Lensfun performs fuzzy matching, but the string must be specific enough to avoid collisions with other lenses from the same maker.
`<mount>`	Yes	Mount identifier. Must exactly match a known mount name in the lensfun database. See Section 6.4 for the correct strings.
`<cropfactor>`	Yes	Crop factor of the sensor the lens was calibrated for. For Fujifilm APS-C use 1.53; for GFX/Hasselblad X use 0.79; for full frame use 1.0.
`<calibration>`	Yes	Container element for all <distortion> and <vignetting> child entries.
`<distortion>`	No	One entry per focal length (primes have one; zooms have several). Model attribute must be "ptlens".
`<vignetting>`	No	One entry per focal length × aperture × focus distance combination. Model attribute must be "pa".

6.4 Mount Name Reference

Mount names are case-sensitive and must be spelled exactly as lensfun expects. Using an incorrect name causes the lens entry to be silently ignored. The following are the correct strings for the sensor presets in this tool:

Camera system	Correct mount name in lensfun
Fujifilm X-mount	Fujifilm X
Fujifilm GFX / Hasselblad X2D	Hasselblad X
Sony E-mount (APS-C and full frame)	Sony E
Nikon F-mount	Nikon F
Canon EF (full frame)	Canon EF

Camera system	Correct mount name in lensfun
Canon EF-S (APS-C)	Canon EF-S
Micro Four Thirds	Micro Four Thirds

Note: If in doubt about the exact mount string for your system, search the lensfun system database files for an existing lens on the same mount and copy the string verbatim.

6.5 Distortion Entries: One Per Focal Length

Lensfun expects at most one `<distortion>` entry per focal length within a lens calibration block. For zooms, one entry per calibrated focal length is used with interpolation in between. For primes there is a single entry. The aperture and focus-distance filters in the converter exist to select the single representative entry that goes into the `<distortion>` block.

```
<distortion model="ptlens" focal="56" a="-0.001234" b="0.005678" c="-0.002345" />
```

Note: Darktable applies only one distortion correction value regardless of aperture or focus distance. The default export values (f/8, infinity focus) are the recommended choice for the vast majority of lenses.

6.6 Vignetting Entries: Full Table Required

Unlike distortion, lensfun does use the full vignetting table and interpolates across aperture and focus distance at render time. The exported `.xml` file always contains vignetting entries for every aperture and distance combination present in the LCP. The sort order is: focal length → aperture → focus distance.

```
<vignetting model="pa" focal="56" aperture="1.2" distance="10000"
            k1="-0.2340" k2="0.0560" k3="-0.0120" />
<vignetting model="pa" focal="56" aperture="2.0" distance="10000"
            k1="-0.1890" k2="0.0430" k3="-0.0098" />
```

6.7 Checking for Existing Entries

Before placing a new file in the user database, search the system database for an existing entry for your lens. If one exists, your user-database entry will supplement or override it depending on whether the identifier strings match exactly.

On Linux, search all system database files:

```
grep -r "XF 56mm" /usr/share/lensfun/
```

If a match is found, note the exact `<maker>`, `<model>`, and `<mount>` strings and use them in your new entry. This ensures lensfun treats your calibration data as a replacement rather than a separate lens.

7. End-to-End Workflow: LCP to darktable

The following is the recommended process for integrating a new lens into darktable using this tool.

Step 1 — Obtain the LCP File

Adobe publishes free LCP files for many lenses. They can also be found inside a Lightroom installation (in the `LensProfiles` subfolder) or generated with the Adobe Lens Profile Creator tool. The file has a `.lcp` extension and is an XML/RDF document.

Step 2 — Open the Tool

Open `LCP_Lens_Correction_Converter.html` in any modern browser. No internet connection is required. The file is fully self-contained.

Step 3 — Load the LCP

Drag-and-drop the .lcp file onto the drop zone or click to browse. Verify the status bar reports a reasonable entry count. A lens with calibration at several focal lengths and apertures typically contains 20–60 entries.

Step 4 — Select the Correct Sensor

Click the sensor button corresponding to the camera body the LCP was calibrated for. Verify no sensor mismatch warning appears. If it does, examine the LCP's SensorFormatFactor and choose the sensor whose crop factor is closest.

Step 5 — Set Filter Values

Leave the aperture filter at f/8 (or the auto-selected default) and the focus distance at infinity. These are darktable's defaults for distortion correction. Only change them if you have a specific reason to calibrate at a different working aperture.

Step 6 — Inspect the Distortion Graph (Optional)

Click 'Show distortion graph'. Verify the ACM and PTLens curves are close throughout the frame, especially in the corners. Significant divergence at the corners indicates a large-distortion lens where the PTLens model fit may be imperfect.

Step 7 — Export the XML

Click ↓ Export lensfun .xml. The browser downloads the file.

Step 8 — Check for an Existing Entry

Search the lensfun system database for an existing entry for your lens (see Section 6.7). If found, update the `<maker>`, `<model>`, and `<mount>` values in the exported XML to match exactly.

Step 9 — Prepare the User Database Directory

Ensure the lensfun user database directory exists. Create it if needed:

```
mkdir -p ~/.local/share/lensfun/          # Linux

mkdir -p ~/Library/Application\ Support/lensfun/db/   # macOS
```

Step 10 — Place the File

Copy the exported `.xml` file into the user database directory. The file is immediately ready to use — no further modification is required if the maker/model/mount values are correct.

Step 11 — Verify in darktable

Restart darktable. Open an image captured with the lens. Navigate to the Lens Correction module. The lens should appear in the lens selector. Apply the correction and inspect the result visually, particularly at the frame corners.

Note: If the lens does not appear: (1) verify the user database directory path for your OS, (2) confirm the <mount> string matches exactly, (3) confirm the file is valid XML with no syntax errors introduced during any manual editing.

8. Merging Data into an Existing Database File

If you already maintain a user database file for the same manufacturer or mount, you may prefer to merge the calibration data rather than create a separate file. This keeps the database tidy but requires careful editing to avoid duplicate lens entries or malformed XML.

1. Open both the exported XML file and the target user database file in a text editor.
2. In the exported file, locate the `<calibration>` block and copy the complete set of `<distortion>` and `<vignetting>` lines.
3. In the target file, search for the `<model>` string matching your lens. If found, paste the copied lines into the existing `<calibration>` block. If not found, paste the entire `<lens>...</lens>` block from the exported file as a new sibling entry.
4. Ensure there is **exactly one** `<lens>` **entry** per lens in the file. Duplicate entries cause unpredictable behaviour.
5. Save the file and restart darktable.

Important: XML is whitespace-sensitive in places and completely unforgiving of mismatched tags, missing quotes, or invalid attribute values. Validate your file after editing by opening it in a browser (which will show a parse error if the XML is malformed) or running: xmllint --noout yourfile.xml

9. Technical Reference

9.1 LCP Parsing

The parser uses the browser's built-in `DOMParser` with `application/xml` mode. Lens metadata is extracted from the `stCamera` namespace (`http://ns.adobe.com/photoshop/1.0/camera-profile`). Calibration entries are located by iterating all `rdf:li` elements and finding their child `rdf:Description` nodes.

Per-entry fields extracted:

- `stCamera:ApertureValue` — APEX aperture value; f-number = 2^(ApertureValue ÷ 2)
- `stCamera:FocusDistance` — focus distance in metres; 10000 = infinity
- `stCamera:FocalLength` — focal length in mm
- `stCamera:RadialDistortParam1/2/3` — ACM distortion k1, k2, k3
- `stCamera:VignetteModelParam1/2/3` — ACM vignette α1, α2, α3

9.2 Conversion Precision

All coefficients are output to 6 decimal places. The Cramér's rule solver checks for a near-zero determinant (< 1e−30) and returns `NaN` if the normal equations are singular, which can occur for entries with zero distortion. These appear as '—' in the table.

9.3 Browser Compatibility

The tool uses standard Web APIs: `FileReader`, `DOMParser`, `Canvas 2D`, and ES2020 JavaScript. It is compatible with all current browsers. No polyfills, plugins, or network access are required.

9.4 Lensfun PTLens Model Convention

The PTLens distortion model as used by lensfun is:

$$r_distorted = r_undistorted \times (1 + c \cdot r + b \cdot r^2 + a \cdot r^3)$$

where r is normalised to R_NORM (half the sensor short side). The sign convention means positive b produces barrel distortion correction (inward radial displacement), and negative b produces pincushion correction.

9.5 Lensfun PA Vignetting Model Convention

Lensfun's polynomial approximation vignetting model computes a per-pixel gain factor:

$$L(r) = 1 + k1 \cdot r^2 + k2 \cdot r^4 + k3 \cdot r^6$$

$$gain = 1 / L(r)$$

where r is normalised to R_MAX (half the sensor diagonal). A positive $k1$ darkens the corners (natural vignetting); negative values would brighten them (rarely seen). The vignette graph plots $-\log_2(L(r))$ in EV, showing the total correction applied at each radius.

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